

I earned my bachelor's degree in Electrical and Electronics Engineering in 2014, followed by seven years of professional experience in the telecommunications industry, where I worked extensively with large-scale 2G, 3G, 4G, and 5G systems. During this period, I became increasingly aware of the value and potential of data generated by these complex systems, which motivated me to pursue a master's degree in computer and information technology. This combination of engineering and computing training has provided me with a strong foundation in quantitative analysis, programming, and applied problem-solving, enabling me to approach complex systems from both theoretical and practical perspectives.

Throughout my master's research, I worked extensively with real-world datasets, computational models, and interdisciplinary challenges, experiences that naturally prepared me for doctoral-level research. Building on this background, my doctoral research interest focuses on developing intelligent decision support systems for advanced semiconductor manufacturing, where data-driven insights can enhance process efficiency, reliability, and operational decision-making.

My primary research interests lie at the intersection of applied machine learning, data analytics, and decision support systems. I am particularly interested in how advanced analytics and AI techniques can be leveraged to extract meaningful insights from complex datasets and support evidence-based decision-making in technology-intensive and industrial domains. These research interests are closely aligned with my academic training and professional experience and continue to motivate my pursuit of deeper methodological and applied research during my graduate studies.

My long-term professional goal is to work in a research-driven role, either in academia, industry, or a research-focused organization, where I can contribute to the development of intelligent, data-informed solutions to complex technological and societal challenges. I aspire to take on leadership

responsibilities that combine technical expertise with strategic thinking. I anticipate challenges such as keeping pace with rapidly evolving technologies, managing interdisciplinary research problems, and balancing theoretical rigor with practical impact. I view these challenges as opportunities for growth and believe that rigorous graduate training will equip me with the skills necessary to address them effectively.

During my graduate studies, I have identified three clear and relevant goals:

1. To develop advanced technical expertise in machine learning, statistical modeling, and computational analysis through coursework and applied research.
2. To produce high-quality research outputs, including thesis work, publications, or technical reports, that demonstrate methodological rigor and real-world relevance.
3. To strengthen professional and research communication skills, enabling me to effectively present complex ideas to both technical and non-technical audiences.

I view my doctoral studies as a critical step in transforming my technical background and research interests into impactful contributions to both academia and industry.